

KASSEL, Germany

WMO: 104380

Lat: 51.296N

Lon: 9.443E

Elev: 158

StdP: 99.44

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-12.2	-9.4	-15.4	1.0	-10.6	-13.1	1.2	-8.2	10.2	6.0	9.2	3.9	2.5	0	0.541

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.3	28.4	18.8	26.6	17.8	25.0	17.1	19.9	26.0	18.8	24.4	18.0	23.2	3.2	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.7	12.9	22.1	16.8	12.2	21.2	15.9	11.5	20.1	57.4	26.1	54.0	24.5	51.1	23.1	23.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.8	7.7	6.8	DB	-12.9	32.2	4.5	1.7	-16.1	33.4	-18.8	34.4	-21.3	35.4	-24.5	36.6	(4)
(5)			WB	-13.2	22.0	4.5	1.0	-16.5	22.7	-19.2	23.3	-21.7	23.8	-25.0	24.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.9	0.3	0.2	4.5	7.8	12.7	15.4	17.8	17.6	13.9	9.7	4.4	2.1		
	DBStd	7.33	5.15	4.75	3.73	3.66	3.75	3.59	3.26	3.19	3.01	3.48	3.62	3.81		
	HDD10.0	1316	300	275	172	86	16	2	0	0	3	48	168	246		
	HDD18.3	3521	558	508	428	317	178	102	52	53	136	268	417	504		
	CDD10.0	924	0	1	3	19	99	164	242	237	120	38	1	1		
	CDD18.3	88	0	0	0	0	3	14	35	32	3	1	0	0		
	CDH23.3	747	0	0	0	0	33	122	299	264	28	1	0	0		
	CDH26.7	154	0	0	0	0	1	20	66	66	1	0	0	0		
(14) Wind	WSAvg	2.9	3.9	3.3	3.2	2.9	2.5	2.5	2.3	2.3	2.5	2.8	3.3	3.6		
(15) Precipitation	PrecAvg	706	54	44	52	50	68	75	69	65	57	52	60	67		
	PrecMax	1087	140	125	109	133	152	171	185	152	137	176	126	124		
	PrecMin	498	3	5	10	9	11	18	15	17	13	11	5	6		
	PrecStd	137	32	29	26	24	32	34	38	31	30	35	25	33		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.4	13.1	17.6	21.1	26.3	29.0	30.7	31.6	26.3	21.3	13.2	11.8		
		MCWB	8.5	9.7	10.5	12.3	16.1	19.7	19.5	19.9	16.7	15.4	10.5	9.8		
	2%	DB	8.7	10.1	13.9	18.7	24.1	26.6	28.6	28.4	23.7	18.2	11.5	10.2		
		MCWB	7.2	7.8	8.9	11.3	14.9	17.9	18.8	18.6	16.7	14.3	9.8	8.6		
	5%	DB	7.6	8.0	11.5	16.4	22.0	24.4	26.6	26.2	21.4	16.2	10.4	8.9		
		MCWB	6.3	6.0	8.0	10.2	14.0	16.7	18.2	17.7	15.8	13.1	8.9	7.7		
	10%	DB	6.4	6.0	9.6	14.2	20.1	22.2	24.8	24.2	19.4	14.5	9.3	7.4		
		MCWB	5.2	4.3	7.0	9.0	13.4	15.8	17.5	17.1	15.0	12.0	7.9	6.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.7	10.2	11.1	13.3	17.2	20.9	21.1	21.2	18.8	16.3	11.3	10.1		
		MCDB	9.9	12.6	15.6	19.4	23.2	27.5	28.0	28.4	22.7	19.9	12.1	11.1		
	2%	WB	7.6	8.3	9.7	12.1	15.9	19.0	19.9	19.5	17.5	14.7	10.2	8.8		
		MCDB	8.5	10.1	13.2	17.3	22.0	24.2	26.2	25.8	21.5	17.4	11.3	9.9		
	5%	WB	6.4	6.2	8.6	10.8	14.8	17.7	19.0	18.5	16.4	13.4	9.2	7.6		
		MCDB	7.4	7.7	11.0	15.1	20.7	22.8	24.7	24.2	20.2	15.6	10.3	8.7		
	10%	WB	5.2	4.6	7.3	9.6	13.7	16.6	17.9	17.7	15.4	12.4	8.0	6.5		
		MCDB	6.4	5.9	9.4	13.4	19.4	21.2	23.3	22.8	18.8	14.2	9.1	7.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.0	5.4	6.7	8.4	9.2	8.7	9.3	9.7	8.2	6.7	4.6	3.8		
		MCDBR	4.6	7.3	10.2	11.9	12.5	12.4	12.8	13.4	11.9	9.0	5.7	4.9		
	5% WB	MCWBR	4.2	5.4	6.2	5.9	5.9	5.5	5.4	5.6	5.7	5.3	4.5	4.4		
		MCWBR	4.6	7.1	9.2	10.3	11.5	11.0	10.9	11.5	10.3	8.0	5.4	5.0		
(40) Clear-Sky Solar Irradiance	taub		0.313	0.338	0.374	0.414	0.418	0.416	0.429	0.423	0.387	0.358	0.337	0.310		
		taud	2.390	2.349	2.271	2.200	2.224	2.251	2.219	2.258	2.342	2.410	2.399	2.385		
	Ebn at Noon		704	781	815	822	837	841	821	805	794	748	664	642		
		Edh at Noon	62	83	108	130	133	131	133	121	100	78	61	53		
(44) All-Sky Solar Radiation	RadAvg		0.70	1.38	2.49	3.98	4.77	5.22	4.95	4.23	3.04	1.78	0.80	0.53		
	RadStd		0.10	0.22	0.34	0.49	0.51	0.48	0.66	0.40	0.42	0.18	0.12	0.06		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (42 neighbors)		+0.32	N/A	N/A	N/A	N/A	N/A	N/A	-120	N/A	N/A

Nomenclature: See separate page